

AN EXPLICIT 32-POINT SET WITH A FREE HALF-TURN AND NO CONVEX 7-GON

ABSTRACT. Subercaseaux, Mackey, Qian and Heule closed the 4-fold symmetric case of the 32-point Erdős–Szekeres problem negatively and listed, as a further problem in their conclusions, the realization of a 2-fold symmetric 32-point set with no convex 7-gon. We exhibit one. The set $W^* = P \cup (-P)$, where P is an explicit list of sixteen integer points of absolute value at most 31, lies in general position, is carried to itself by the half-turn $h(z) = -z$, which acts on it freely with sixteen orbits of size two, and contains no seven points in convex position; its largest subsets in convex position have size 6, and there are 56261 of them. Realizability is immediate because the points are integers. Combined with the negative 4-fold result and an orbit argument, this determines which rotational orders can act on a 7-gon-free set of 32 points: only $s \in \{1, 2, 4\}$ are possible, $s = 1$ and $s = 2$ occur, and $s = 4$ does not. Nothing here bounds $g(7)$.

1. THE PROBLEM

Write $g(k)$ for the least n such that every set of n points in general position in the plane (no three collinear) contains k points in convex position. Erdős and Szekeres proved $g(k) \geq 2^{k-2} + 1$ and conjectured equality [1, 2]; the only known values are $g(4) = 5$, $g(5) = 9$ and $g(6) = 17$ [4], so $g(7)$ is open and $32 = 2^{7-2}$ is the conjectured extremal size, not a proved one.

Subercaseaux, Mackey, Qian and Heule [5] settle the case $n = 32$, $k = 7$ under a 4-fold rotational symmetry assumption negatively, and their Conclusions name what is left. Quoted from the e-print:

Further problems in this line of work include finding a realization of a 2-fold symmetric construction of 32 points without a convex 7-gon, and designing an efficient solver that can handle collinearity constraints for the realizability problem.

The sentence carries no number: it is the further-problems sentence of a Conclusions section, and it is reproduced verbatim so that what is answered below does not depend on a label. The ambient definitions are those of the source. General position is no three points collinear; a set S in general position is in convex position if and only if every 4-point subset of S is, equivalently if and only if no point of S lies in the convex hull of three

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others, a consequence of Carathéodory's theorem. Here the exhibited object is a list of integers, so realizability needs no appeal to the existential theory of the reals.

Constructions of 2^{k-2} -point k -gon-free sets, generalizing the previously known families, are given by Baek and Balko [6]; symmetry is not considered there. No priority or novelty is claimed for the set below.

2. THE CONSTRUCTION

Let P be the sixteen integer points

$$\begin{array}{cccc} (18, -4) & (6, -18) & (-17, 4) & (-3, -18) \\ (-5, 20) & (12, 21) & (15, 23) & (-13, 5) \\ (20, -25) & (-22, 2) & (23, 27) & (-22, -26) \\ (23, -2) & (-30, -31) & (27, -1) & (4, 19) \end{array}$$

and put $W^* = P \cup (-P)$, that is, the thirty-two points

$$\begin{array}{cccc} (18, -4) & (-18, 4) & (6, -18) & (-6, 18) \\ (-17, 4) & (17, -4) & (-3, -18) & (3, 18) \\ (-5, 20) & (5, -20) & (12, 21) & (-12, -21) \\ (15, 23) & (-15, -23) & (-13, 5) & (13, -5) \\ (20, -25) & (-20, 25) & (-22, 2) & (22, -2) \\ (23, 27) & (-23, -27) & (-22, -26) & (22, 26) \\ (23, -2) & (-23, 2) & (-30, -31) & (30, 31) \\ (27, -1) & (-27, 1) & (4, 19) & (-4, -19) \end{array}$$

Every coordinate is an integer of absolute value at most 31, the bound being attained at $(-30, -31)$. That bound is not claimed to be minimal.

Theorem 1. *W^* is a set of 32 points in general position in the plane; the half-turn $h(z) = -z$ satisfies $h(W^*) = W^*$, has order 2, and acts on W^* freely, with exactly 16 orbits of size 2; and W^* contains no 7 points in convex position. Its largest subsets in convex position have size exactly 6, and there are 56261 of them.*

Thus a realization of the object named in the sentence quoted above exists.

Proof. The list has 32 entries and they are pairwise distinct, so $|W^*| = 32$; the origin does not occur in the list. The map h is linear, hence an affine bijection of the plane, so it preserves collinearity, convex hulls and convex position; $h(W^*) = W^*$ holds entry by entry by construction of W^* as $P \cup (-P)$; $h \circ h = \text{id}$ and $h \neq \text{id}$, so h has order exactly 2. The unique fixed point of h in the plane is the origin, which is not in W^* ; therefore no point of W^* is fixed, every orbit has size 2, and there are $32/2 = 16$ of them.

General position and the absence of seven points in convex position are finite decisions, and no short argument for them is claimed; they were computed as described in §4. All $\binom{32}{3} = 4960$ orientation determinants $\det(b - a, c - a)$ are nonzero, their absolute values ranging from 1 to 2892.

For the second part, the subsets of W^* in convex position, counted by size, are

size	3	4	5	6	≥ 7
number in convex position	4960	24322	59002	56261	0
$\binom{32}{\text{size}}$	4960	35960	201376	906192	

so the maximum size of a subset in convex position is 6 and the number of those is 56261. Convex position is inherited by subsets, so “no 7 in convex position” already gives “no m in convex position” for every $m \geq 7$. $\square$

Remark. Two entries of the table are forced: every triple of a set in general position is in convex position, which fixes the size-3 column, and W^* must contain a convex hexagon since $g(6) = 17 \leq 32$. What is not forced is the 0 in the final column, that is, that the maximum is *exactly* 6. An explicit hexagon, for a reader who wants one to hand, is

$$(18, -4) \quad (-18, 4) \quad (6, -18) \quad (-6, 18) \quad (-3, -18) \quad (3, 18).$$

3. ROTATION ORDERS AT $n = 32$

Proposition 2. *Let W be a set of 32 points in general position with no 7 points in convex position, and let ρ be a rotation of the plane of order s with $\rho(W) = W$. Then $s \in \{1, 2, 4\}$.*

Proof. The cyclic group $\langle \rho \rangle$ has order s and acts on W ; the only point of the plane with nontrivial stabiliser is the centre of ρ , so every orbit has size s except for at most one orbit of size 1. Hence $32 \equiv 0$ or $1 \pmod{s}$, i.e. s divides 32 or s divides 31, which leaves $s \in \{1, 2, 4, 8, 16, 31, 32\}$. In particular $s \in \{3, 5, 6\}$ is impossible, since $32 \equiv 2$ modulo each of 3, 5 and 6. Finally, if $s \geq 7$ then W contains an orbit of size s , which is the vertex set of a regular s -gon; any 7 of its vertices are in convex position, a contradiction. This deletes 8, 16, 31, 32 and leaves $\{1, 2, 4\}$. $\square$

Corollary 3. *For $n = 32$ and $k = 7$: a set with $s = 1$ exists (Erdős–Szekeres [2]); a set with $s = 2$ exists (Theorem 1); no set with $s = 4$ exists ([5], established there by an exhaustive computation we do not reproduce); and no other rotational order is possible (Proposition 2).*

Of the four lines, only the $s = 2$ line is new here; the $s = 4$ line is the theorem of [5] and the $s = 1$ line is classical. The corollary concerns rotations only: the mirror-symmetric case at $n = 32$ is untouched. Nor does anything above bound $g(7)$, which is unknown; the inequality $g(7) \geq 33$ is classical, from the asymmetric 2^{k-2} construction of [2] (see also [3, Thm. 2.6]), and W^* does not improve it.

4. VERIFICATION

The finite parts of Theorem 1 were recomputed by two structurally different exhaustive procedures, in exact integer arithmetic with no floating-point decision anywhere. The first applies the Carathéodory criterion of §1 to each

of the $\binom{32}{7} = 3,365,856$ seven-subsets, with no pruning, and finds 0 in convex position. The second never mentions the number 7: it enumerates every subset in convex position *at every size at once*, by growing angularly ordered left-turning chains from each vertex in turn with the closing turn checked explicitly, and so returns the maximum directly; because that enumeration is untruncated, its answer 6 settles all sizes above 6 simultaneously. The two agree on the size-4 column, where both are cheap: 24322 of $\binom{32}{4} = 35960$. The criterion was pinned in both polarities before use, so that neither answer is vacuous.

A program that carries out all of this on the integers printed in §2, together with a recorded run of it, accompanies this note; it is not needed in order to redo the work, since every input it consumes is printed above. It does *not* re-run the search that produced W^* , which is why W^* is pinned as a printed list of integers, and it does *not* recompute the 4-fold negative result quoted in Corollary 3, which is taken from [5].

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